Swift REST API supported operations

StorageGRID 11.7

NetApp
March 05, 2024

This PDF was generated from https://docs.netapp.com/us-en/storagegrid-117/swift/supported-swift-api-endpoints.html on March 05, 2024. Always check docs.netapp.com for the latest.

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Swift REST API supported operations

The StorageGRID system supports most operations in the OpenStack Swift API. Before integrating Swift REST API clients with StorageGRID, review the implementation details for account, container, and object operations.

Operations supported in StorageGRID

The following Swift API operations are supported:

- Account operations
- Container operations
- Object operations

Common response headers for all operations

The StorageGRID system implements all common headers for supported operations as defined by the OpenStack Swift Object Storage API v1.

Related information

OpenStack: Object Storage API

Supported Swift API endpoints

StorageGRID supports the following Swift API endpoints: the info URL, the auth URL, and the storage URL.

info URL

You can determine the capabilities and limitations of the StorageGRID Swift implementation by issuing a GET request to the Swift base URL with the /info path.

```
https://FQDN | Node IP:Swift Port/info/
```

In the request:

- *FQDN* is the fully qualified domain name.
- *Node IP* is the IP address for the Storage Node or the Gateway Node on the StorageGRID network.
- *Swift Port* is the port number used for Swift API connections on the Storage Node or Gateway Node.

For example, the following info URL would request information from a Storage Node with the IP address of 10.99.106.103 and using port 18083.

```
https://10.99.106.103:18083/info/
```

The response includes the capabilities of the Swift implementation as a JSON dictionary. A client tool can parse the JSON response to determine the capabilities of the implementation and use them as constraints for subsequent storage operations.

The StorageGRID implementation of Swift allows unauthenticated access to the info URL.

auth URL

A client can use the Swift auth URL to authenticate as a tenant account user.

```
https://FQDN | Node IP:Swift Port/auth/v1.0/
```

You must provide the tenant account ID, user name, and password as parameters in the `X-Auth-User` and `X-Auth-Key` request headers, as follows:

```
X-Auth-User: Tenant_Account_ID:Username

X-Auth-Key: Password
```

In the request headers:

- *`Tenant_Account_ID`* is the account ID assigned by StorageGRID when the Swift tenant was created. This is the same tenant account ID used on the Tenant Manager sign-in page.
- *`Username`* is the name of a tenant user that has been created in the Tenant Manager. This user must belong to a group that has the Swift Administrator permission. The tenant's root user can't be configured to use the Swift REST API.

  If Identity Federation is enabled for the tenant account, provide the username and password of the federated user from the LDAP server. Alternatively, provide the LDAP user's domain name. For example:

  ```
  X-Auth-User: Tenant_Account_ID:Username@Domain_Name
  ```

- *`Password`* is the password for the tenant user. User passwords are created and managed in the Tenant Manager.

The response to a successful authentication request returns a storage URL and an auth token, as follows:

```
X-Storage-Url: https://FQDN | Node_IP:Swift_Port/v1/Tenant_Account_ID

X-Auth-Token: token

X-Storage-Token: token
```

By default, the token is valid for 24 hours from generation time.

Tokens are generated for a specific tenant account. A valid token for one account does not authorize a user to access another account.

storage URL

A client application can issue Swift REST API calls to perform supported account, container, and object operations against a Gateway Node or Storage Node. Storage requests are addressed to the storage URL returned in the authentication response. The request must also include the X-Auth-Token header and value returned from the auth request.

```
https://FQDN | IP:Swift_Port/v1/Tenant_Account_ID
```

`[/container][/object]`

`X-Auth-Token: token`

Some storage response headers that contain usage statistics might not reflect accurate numbers for recently modified objects. It might take a few minutes for accurate numbers to appear in these headers.

The following response headers for account and container operations are examples of those that contain usage statistics:

- `X-Account-Bytes-Used`
- `X-Account-Object-Count`
- `X-Container-Bytes-Used`
- `X-Container-Object-Count`

Related information

Configure tenant accounts and connections

Account operations

Container operations

Object operations

Account operations

The following Swift API operations are performed on accounts.

GET account

This operation retrieves the container list associated with the account and account usage statistics.

The following request parameter is required:

- `Account`

The following request header is required:

- `X-Auth-Token`

The following supported request query parameters are optional:

- `Delimiter`
- `End_marker`
- `Format`
- `Limit`
- `Marker`
- `Prefix`

A successful execution returns the following headers with an “HTTP/1.1 204 No Content” response if the account is found and has no containers or the container list is empty; or an “HTTP/1.1 200 OK” response if the account is found and the container list is not empty:

- `Accept-Ranges`
- `Content-Length`
- `Content-Type`
- `Date`
- `X-Account-Bytes-Used`
- `X-Account-Container-Count`
- `X-Account-Object-Count`
- `X-Timestamp`
- `X-Trans-Id`

HEAD account

This operation retrieves account information and statistics from a Swift account.

The following request parameter is required:

- `Account`

The following request header is required:

- `X-Auth-Token`

A successful execution returns the following headers with an “HTTP/1.1 204 No Content” response:

- `Accept-Ranges`
- `Content-Length`
- `Date`
- `X-Account-Bytes-Used`
- `X-Account-Container-Count`
- `X-Account-Object-Count`
- `X-Timestamp`
- `X-Trans-Id`

Related information

Monitor and audit operations

Container operations

StorageGRID supports a maximum of 1,000 containers per Swift account. The following

Swift API operations are performed on containers.

DELETE container

This operation removes an empty container from a Swift account in a StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`

The following request header is required:

- `X-Auth-Token`

A successful execution returns the following headers with an "HTTP/1.1 204 No Content" response:

- `Content-Length`
- `Content-Type`
- `Date`
- `X-Trans-Id`

GET container

This operation retrieves the object list associated with the container along with container statistics and metadata in a StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`

The following request header is required:

- `X-Auth-Token`

The following supported request query parameters are optional:

- `Delimiter`
- `End_marker`
- `Format`
- `Limit`
- `Marker`
- `Path`
- `Prefix`

A successful execution returns the following headers with an "HTTP/1.1 200 Success" or a "HTTP/1.1 204 No

Content" response:

- `Accept-Ranges`
- `Content-Length`
- `Content-Type`
- `Date`
- `X-Container-Bytes-Used`
- `X-Container-Object-Count`
- `X-Timestamp`
- `X-Trans-Id`

HEAD container

This operation retrieves container statistics and metadata from a StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`

The following request header is required:

- `X-Auth-Token`

A successful execution returns the following headers with an "HTTP/1.1 204 No Content" response:

- `Accept-Ranges`
- `Content-Length`
- `Date`
- `X-Container-Bytes-Used`
- `X-Container-Object-Count`
- `X-Timestamp`
- `X-Trans-Id`

PUT container

This operation creates a container for an account in a StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`

The following request header is required:

- `X-Auth-Token`

A successful execution returns the following headers with an "HTTP/1.1 201 Created" or "HTTP/1.1 202 Accepted" (if the container already exists under this account) response:

- `Content-Length`
- `Date`
- `X-Timestamp`
- `X-Trans-Id`

A container name must be unique in the StorageGRID namespace. If the container exists under another account, the following header is returned: "HTTP/1.1 409 Conflict."

Related information

Monitor and audit operations

Object operations

The following Swift API operations are performed on objects. These operations can be tracked in the StorageGRID audit log.

DELETE object

This operation deletes an object's content and metadata from the StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`
- `Object`

The following request header is required:

- `X-Auth-Token`

A successful execution returns the following response headers with an `HTTP/1.1 204 No Content` response:

- `Content-Length`
- `Content-Type`
- `Date`
- `X-Trans-Id`

When processing a DELETE Object request, StorageGRID attempts to immediately remove all copies of the object from all stored locations. If successful, StorageGRID returns a response to the client immediately. If all copies can't be removed within 30 seconds (for example, because a location is temporarily unavailable), StorageGRID queues the copies for removal and then indicates success to the client.

For more information, see How objects are deleted.

GET object

This operation retrieves the object content and gets the object metadata from a StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`
- `Object`

The following request header is required:

- `X-Auth-Token`

The following request headers are optional:

- `Accept-Encoding`
- `If-Match`
- `If-Modified-Since`
- `If-None-Match`
- `If-Unmodified-Since`
- `Range`

A successful execution returns the following headers with an `HTTP/1.1 200 OK` response:

- `Accept-Ranges`
- `Content-Disposition`, returned only if `Content-Disposition` metadata was set
- `Content-Encoding`, returned only if `Content-Encoding` metadata was set
- `Content-Length`
- `Content-Type`
- `Date`
- `ETag`
- `Last-Modified`
- `X-Timestamp`
- `X-Trans-Id`

HEAD object

This operation retrieves metadata and properties of an ingested object from a StorageGRID system.

The following request parameters are required:

- `Account`
- `Container`
- `Object`

The following request header is required:

- `X-Auth-Token`

A successful execution returns the following headers with an "HTTP/1.1 200 OK" response:

- `Accept-Ranges`
- `Content-Disposition`, returned only if `Content-Disposition` metadata was set
- `Content-Encoding`, returned only if `Content-Encoding` metadata was set
- `Content-Length`
- `Content-Type`
- `Date`
- `ETag`
- `Last-Modified`
- `X-Timestamp`
- `X-Trans-Id`

PUT object

This operation creates a new object with data and metadata, or replaces an existing object with data and metadata in a StorageGRID system.

StorageGRID supports objects up to 5 TiB (5,497,558,138,880 bytes) in size.

> Conflicting client requests, such as two clients writing to the same key, are resolved on a "latest-wins" basis. The timing for the "latest-wins" evaluation is based on when the StorageGRID system completes a given request, and not on when Swift clients begin an operation.

The following request parameters are required:

- `Account`
- `Container`
- `Object`

The following request header is required:

- `X-Auth-Token`

The following request headers are optional:

- `Content-Disposition`

- `Content-Encoding`

  Don't use chunked `Content-Encoding` if the ILM rule that applies to an object filters objects based on size and uses synchronous placement on ingest (the Balanced or Strict options for Ingest Behavior).

- `Transfer-Encoding`

  Don't use compressed or chunked `Transfer-Encoding` if the ILM rule that applies to an object filters objects based on size and uses synchronous placement on ingest (the Balanced or Strict options for Ingest Behavior).

- `Content-Length`

  If an ILM rule filters objects by size and uses synchronous placement on ingest, you must specify `Content-Length`.

  > If you don't follow these guidelines for `Content-Encoding`, `Transfer-Encoding`, and `Content-Length`, StorageGRID must save the object before it can determine object size and apply the ILM rule. In other words, StorageGRID must default to creating interim copies of an object on ingest. That is, StorageGRID must use the Dual Commit option for Ingest Behavior.

  For more information about synchronous placement and ILM rules, see Data-protection options for ingest.

- `Content-Type`
- `ETag`
- `X-Object-Meta-<name\>` (object-related metadata)

  If you want to use the **User defined creation time** option as the Reference time for an ILM rule, you must store the value in a user-defined header named `X-Object-Meta-Creation-Time`. For example:

  ```
  X-Object-Meta-Creation-Time: 1443399726
  ```

  This field is evaluated as seconds since January 1, 1970.

- `X-Storage-Class: reduced_redundancy`

  This header affects how many object copies StorageGRID creates if the ILM rule that matches an ingested object specifies an Ingest Behavior of Dual Commit or Balanced.

  - **Dual commit**: If the ILM rule specifies the Dual commit option for Ingest Behavior, StorageGRID creates a single interim copy as the object is ingested (single commit).
  - **Balanced**: If the ILM rule specifies the Balanced option, StorageGRID makes a single interim copy only if the system can't immediately make all copies specified in the rule. If StorageGRID can perform synchronous placement, this header has no effect.

    The `reduced_redundancy` header is best used when the ILM rule that matches the object creates a single replicated copy. In this case using `reduced_redundancy` eliminates the unnecessary creation and deletion of an extra object copy for every ingest operation.

Using the `reduced_redundancy` header is not recommended in other circumstances because it increases the risk the loss of object data during ingest. For example, you might lose data if the single copy is initially stored on a Storage Node that fails before ILM evaluation can occur.

> Having only one replicated copy for any time period puts data at risk of permanent loss. If only one replicated copy of an object exists, that object is lost if a Storage Node fails or has a significant error. You also temporarily lose access to the object during maintenance procedures such as upgrades.

Note that specifying `reduced_redundancy` only affects how many copies are created when an object is first ingested. It does not affect how many copies of the object are made when the object is evaluated by the active ILM policy and does not result in data being stored at lower levels of redundancy in the StorageGRID system.

A successful execution returns the following headers with an "HTTP/1.1 201 Created" response:

- `Content-Length`
- `Content-Type`
- `Date`
- `ETag`
- `Last-Modified`
- `X-Trans-Id`

OPTIONS request

The OPTIONS request checks the availability of an individual Swift service. The OPTIONS request is processed by the Storage Node or Gateway Node specified in the URL.

OPTIONS method

For example, client applications can issue an OPTIONS request to the Swift port on a Storage Node, without providing Swift authentication credentials, to determine whether the Storage Node is available. You can use this request for monitoring or to allow external load balancers to identify when a Storage Node is down.

When used with the info URL or the storage URL, the OPTIONS method returns a list of supported verbs for the given URL (for example, HEAD, GET, OPTIONS, and PUT). The OPTIONS method can’t be used with the auth URL.

The following request parameter is required:

- `Account`

The following request parameters are optional:

- `Container`
- `Object`

A successful execution returns the following headers with an “HTTP/1.1 204 No Content” response. The OPTIONS request to the storage URL does not require that the target exists.

- `Allow` (a list of supported verbs for the given URL, for example, HEAD, GET, OPTIONS, and PUT)
- `Content-Length`
- `Content-Type`
- `Date`
- `X-Trans-Id`

Related information

Supported Swift API endpoints

Error responses to Swift API operations

Understanding the possible error responses can help you troubleshoot operations.

The following HTTP status codes might be returned when errors occur during an operation:

Swift error name	HTTP status
AccountNameTooLong, ContainerNameTooLong, HeaderTooBig, InvalidContainerName, InvalidRequest, InvalidURI, MetadataNameTooLong, MetadataValueTooBig, MissingSecurityHeader, ObjectNameTooLong, TooManyContainers, TooManyMetadataItems, TotalMetadataTooLarge	400 Bad Request
AccessDenied	403 Forbidden
ContainerNotEmpty, ContainerAlreadyExists	409 Conflict
InternalError	500 Internal Server Error
InvalidRange	416 Requested Range Not Satisfiable
MethodNotAllowed	405 Method Not Allowed
MissingContentLength	411 Length Required
NotFound	404 Not Found
NotImplemented	501 Not Implemented
PreconditionFailed	412 Precondition Failed
ResourceNotFound	404 Not Found

Swift error name	HTTP status
Unauthorized	401 Unauthorized
UnprocessableEntity	422 Unprocessable Entity

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